

6. (a) e.g. E-field, force independent of speed, B-field, force $\propto$ speed ... B2
 E-field, force along field direction, B-field, force normal etc ... B2 [4]
- (b) (i) out of plane of paper (not 'upwards')..... B1
 (ii) $mv^2 / r = Bqv$ C1
 $r = (1.67 \times 10^{-27} \times 4.5 \times 10^6) / (0.12 \times 1.6 \times 10^{-19})$ C1
 $r = 0.39 \text{ m}$ A1 [4]
- (c) (i) arrow pointing up page B1
 (ii) $Bqv = Eq$ C1
 $E = 0.12 \times 4.5 \times 10^6$
 $= 5.4 \times 10^5 \text{ V m}^{-1}$ A1 [3]
- (d) gravitational force $\ll F_B$ or F_E B1 [1]